

10-Mark Questions & Answers — DBMS Module 2: Relational Algebra & Calculus

Q1. Explain Relational Algebra operations (Fundamental & Derived), Joins (Inner, Outer, Semi, Anti), and Relational Calculus in detail with schema queries.

Comprehensive solution covering:

1. Six Fundamental Relational Algebra Operations (Selection, Projection, Union, Set Difference, Cartesian Product, Rename).
2. Derived Operations: Intersection, Natural Join, Theta Join, Division operator ($\div$ for "for all" queries).
3. Left, Right, and Full Outer Joins with step-by-step table examples.
4. Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC) with existential ($\exists$) and universal ($\forall$) quantifiers.
5. Equivalence between SQL SELECT queries and Relational Algebra expressions.